

NINGNING ZHANG

Pricing & Decision Intelligence | Forecasting • Choice • Revenue Optimization

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Ph.D. in Statistics from The University of Hong Kong, working on airline revenue management and dynamic pricing. I connect demand forecasting, customer price response, and constrained optimization into testable, explainable pricing systems.

EXPERIENCE

- Postdoctoral Researcher | TravelSky Technology LimitedSep 2025 - Present
 - Lead core algorithm design and POC validation for the forecast-optimize-control loop in airline revenue management and dynamic pricing.
 - Connect demand, capacity, and pricing through hierarchical Bayesian forecasting, DLP / bid price, Bellman DP, customer choice, and WTP modeling.
 - Built a reproducible simulation benchmark; the new stack achieved a 56.8% simulated revenue lift in a fixed-seed stress scenario.
- Postdoctoral Researcher | The University of Hong KongJan 2025 - Aug 2025
 - Research in financial time series, conditional moments, statistical modeling, and uncertainty analysis.
- Algorithm & Data Analytics Intern | Meituan • DiDi2017 - 2018
 - Delivery-time feature engineering, supply-demand analysis, pricing evaluation, and experiment segmentation.

SELECTED PROJECTS & EVIDENCE

- Airline RM Simulation & Algorithm Evaluation Platform2026
 - Seven airports, 12 directed legs, and 42 O&D; markets; common random numbers ensure identical passenger streams.
 - Bayesian online forecasting + DLP + Bellman DP delivered a 56.8% simulated lift over historical mean + EMSR-b; Oracle network control delivered 39.24%.
- Dynamic Pricing & Customer Choice Engine2026
 - Unified customer choice, WTP, purchase probability, and opportunity cost in a constrained revenue-maximization framework.
- Pricing & Revenue Optimization AI Copilot2026
 - LLM tool calling for decision traceability, what-if re-optimization, diagnostics, and constraint checks.

EDUCATION

- Ph.D. in Statistics | The University of Hong Kong2019 - 2024
 - Full Ph.D. scholarship; research in time-series analysis and forecasting.
- M.S. in Statistics | Beijing Jiaotong University2016 - 2019
 - GPA 93.51 / 100; ranked 1st in the program.
- B.S. in Information and Computing Science | Beijing Jiaotong University2012 - 2016

TECH STACK

Modeling: Python • Bayesian Time Series • LightGBM • Choice Modeling • WTP
Optimization: Linear Programming • DLP • Bid Price • Bellman DP
Systems & Evaluation: Streamlit • Backtesting • Common Random Numbers • LLM Tool Calling

* The 56.8% and 39.24% figures are simulation results, not online production lifts.